Order-Dependent Flaky Tests in Java Projects

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Flaky tests pass and fail on the same code without any change to the program under test. This study reruns each test of 26 open-source Java projects in shuffled orders and records every outcome together with the order in which the tests ran.

Two reports to the Quill project describe the first flaky tests that the study found.
https://github.com/quillbuild/quill/issues/412
https://github.com/quillbuild/quill/issues/518

The reruns use the Tessera runner [1] on the Harbor service [2], and the scripts are archived [3].

References

[1] The Tessera developers. Tessera, a test runner for Java.
https://github.com/tessera-tools/tessera

[2] The Harbor developers. Harbor 2.1, a build service.
https://github.com/harbor-ci/harbor/releases/tag/v2.1

[3] Anonymous. Scripts of the study of order-dependent flaky tests.
https://github.com/brandt-lab/order-flakes

Data Availability

The recorded outcomes are at https://drive.google.com/drive/folders/1AbCdEfGhIjKlMnOpQrStUv.